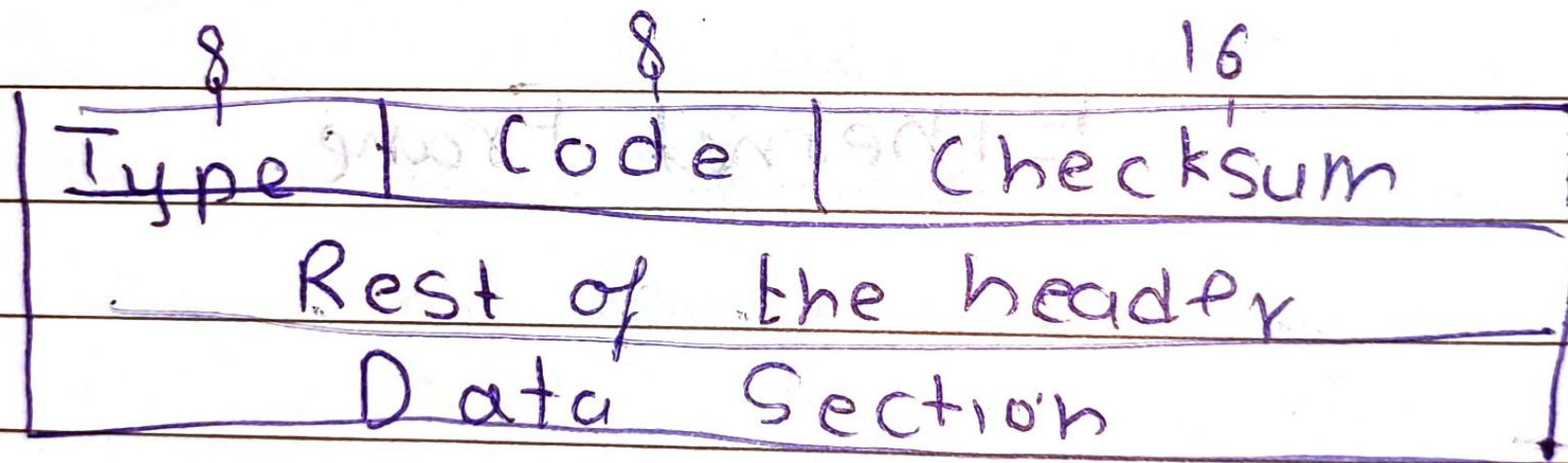


* ICMP (Internet Control Message Protocol)

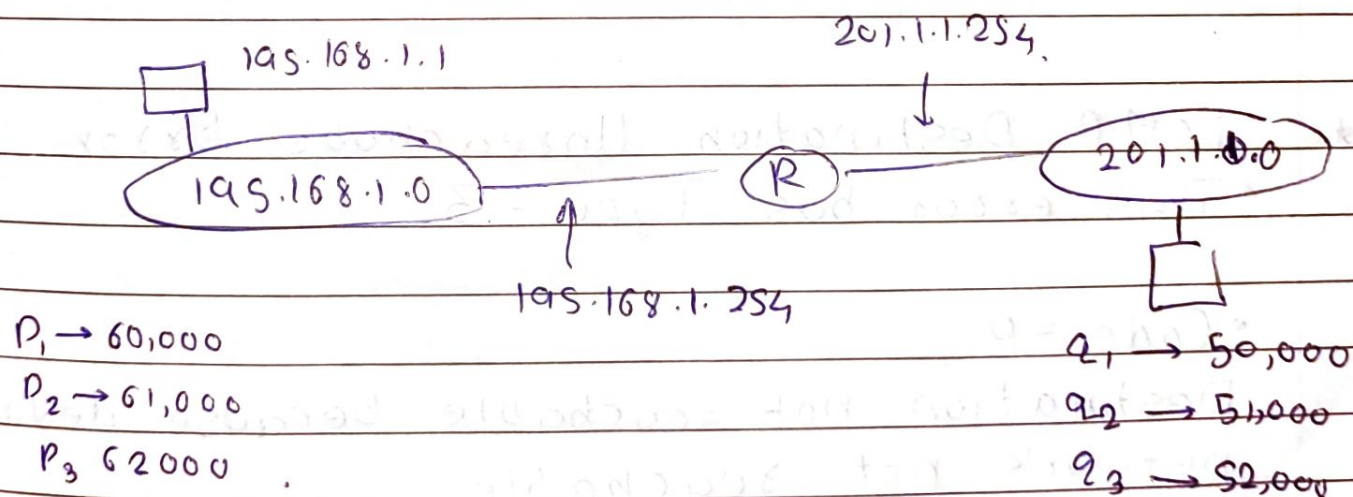
- Used to report about error's
- It is also used to query the network to obtain certain information



ICMP Packet format

- After checksum is calculated pseud header is not part of UDP packet
- At receiver pseudo header is regenerated based on the received data.

- In case of errors "Rest of the headers" is set to all zeros. (Error Reporting)
- For error reporting data section contains original IP header + 8 bytes of the data, (i.e. Source Port, dest port, total length, checksum).
- IP, ARP, UDP, TCP can make calls to ICMP based on specific situations, such as network failure, host failure, port not available.



- ARP request is sent by 195.168.1.1 to R to obtain physical address
- The IP packet is transported to R
- Now if 201.1.1.0 network is down (Not functioning.)

60 00 0	50 00 0
12	CSUM
n	

UDP Packet

4	5	00	32
101		000	0
2	17	HCSUM	
195.168.1.1			
201.1.1.1			
Payload.			

IP Packet

PRI	SFD	DPA	SPA	T/L	Payload	CRC
-----	-----	-----	-----	-----	---------	-----

Ethernet Frame

★ 20 ICMP Destination Unreachable Error

• This error has Type = 3.

• Code = 0

Destination not reachable because destination network not reachable.

• Code = 1

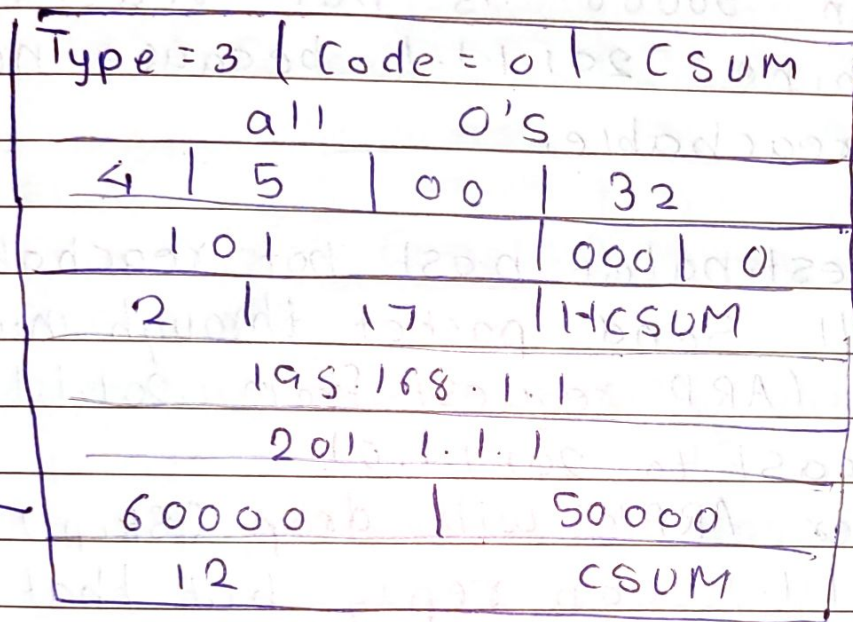
Destination not reachable because destination host is not reachable.

• Code = 2 ← destination not reachable because destination protocol (TCP/UDP) not reachable.

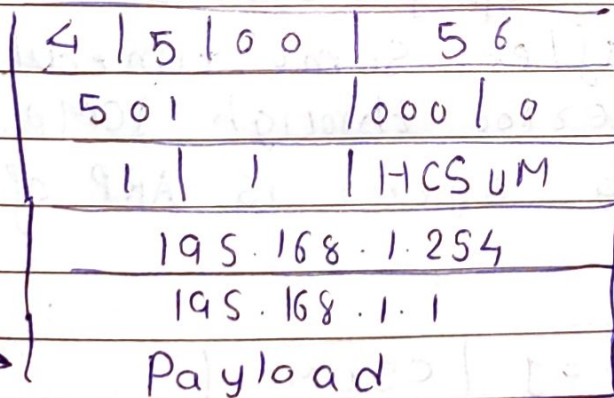
• Code = 3

Destination not reachable because
destination process (Port) is not reachable

*Case 1: Destination network not reachable



ICMP Packet



IP Packet

[PRE | S/D | DPA | SPA | T/L | Payload | CRC]
Ethernet Frame

The ICMP packet will then reach 195.168.1.1
• First the ethernet frame will reach, R1 will check and send IP packet to IP, IP will check protocol field and send the payload (ICMP packet) to ICMP.

• The ICMP will then parse
- Destination 50000 is not reachable on machine 201.1.1.1 because network is not reachable.

* Case 2: Destination host not reachable

• Router will send packet through interface 201.1.1.254 (ARP request from 201.1.1.254 to Broadcast to 201.1.1.0).

• All other ARP's will drop (Skip).

Only 201.1.1.1 can reply but that host is down too.

• As there is no reply from 201.1.1.1 to ARP request, after some timeout.

• R will send error through ICMP.

• Here reporting agent is ARP of router R.

Type = 3	Code = 1	Checksum
all 0's		

rest All same

* Case 3: Destination protocol not reachable

• Reporting agent: IP of 201.1.1.1

• Type = 3, Code = 2.

[2 | 1 | 1 | HCsum]

Only TTL field of IP packet will change as there will be increase in hop as reporting agent is IP of 201.1.1.1

Source IP will change to 201.1.1.1

Rest Same

* Case 4:

- Type : 3 , Code = 3
- Rest all Same
- Reporting Agent: UDP of 201.1.1.1

* How is process found based on port

Process Id	Port
q1	50,000
⋮	⋮

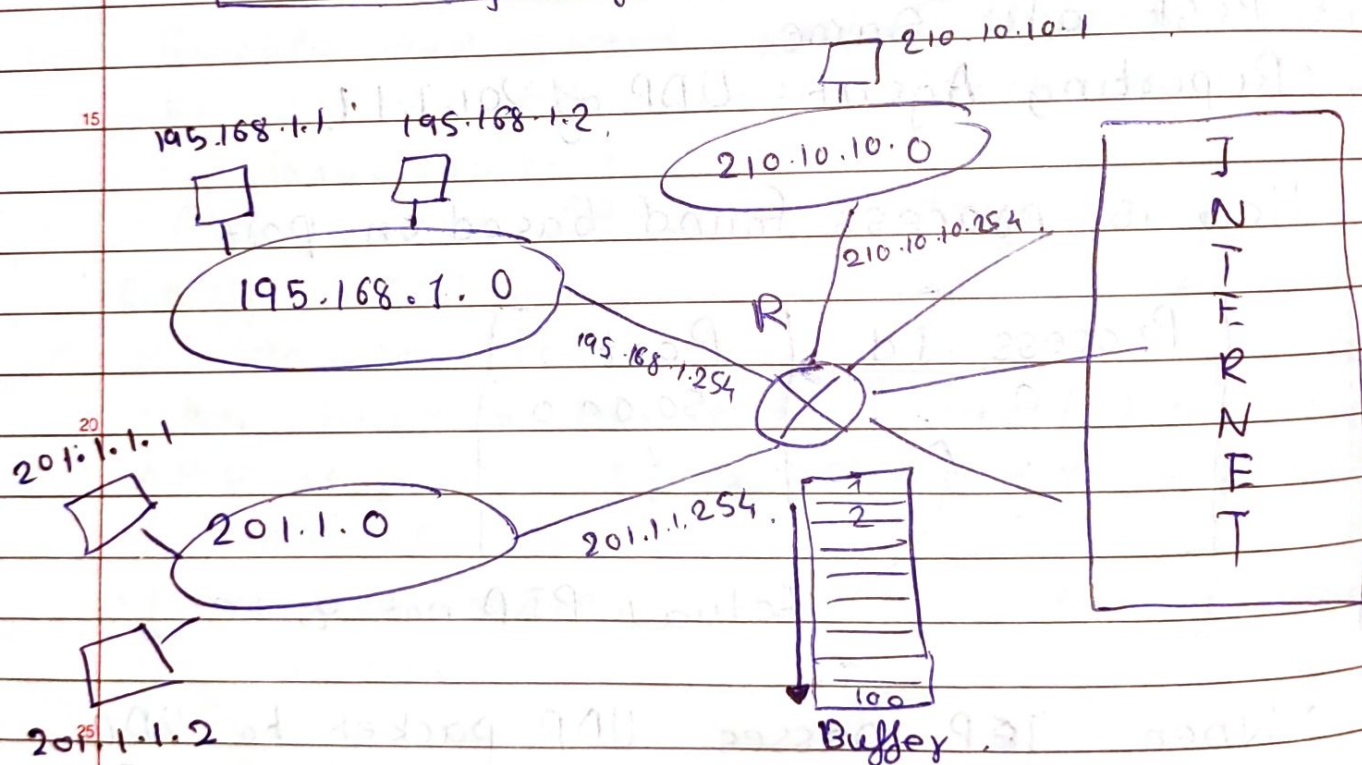
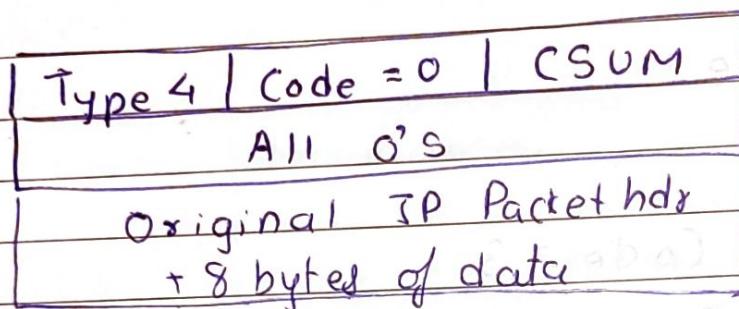
Actual PID not q1

• When IP passes UDP packet to UDP, UDP checks the above table and finds process.

• If process is not running on port 50,000, UDP of 201.1.1.1 makes an ICMP call

* Source Quench Problem/Error.

- When buffer space of router is exhausted it asks the further IP packet sources to quench (slow down). Rate of generation of
- Reporting agent: Router (IP of Router).
- Type: 4.



- Assume all machines are sending series of packets to R. and buffer of R is full
- Lets say process P on 195.168.1.1 on port 6000 is taking number n and process Q, on 210.10.10.1 is finding n^2 on port 50,000

UDP Packet

60000	50000
12	CSUM
n	

4	5	00	32
5		1000	10
6	17	HCSUM	
195.168.1.1			
210.10.10.1			

UDP Packet
IP packet

PRE | SFD | DPA | SPA | T/L | Payload | CRC
Ethernet Frame.

The packet will reach IP of R but the buffer is ~~slow~~ full, So R tells 195.168.1.1 to slow down.

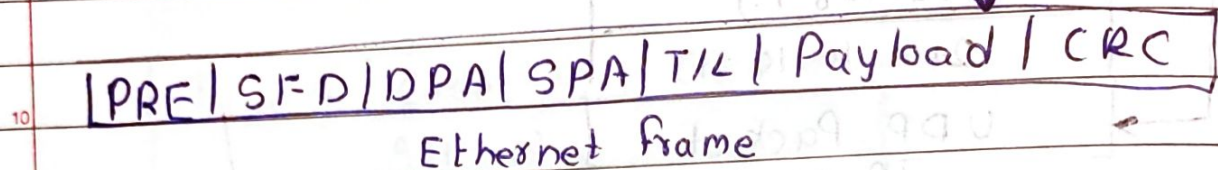
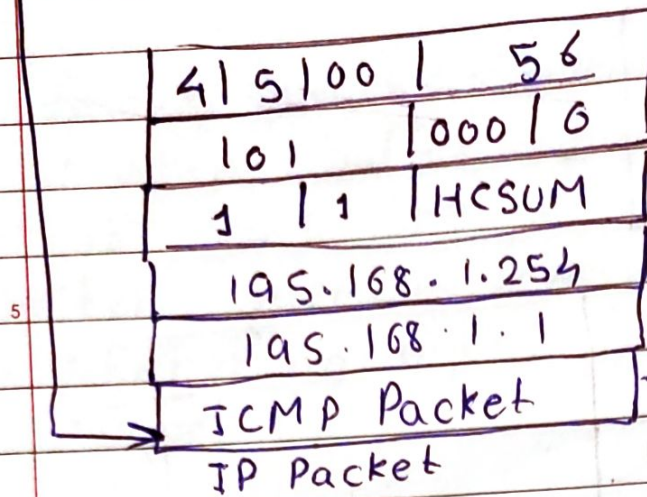
Type = 4	Code = 0	CSUM
All 0's		
4	5	00 32
5	100010	
6	17	HCSUM
195.168.1.1		
210.10.10.1		
60000	50000	
12	CSUM	

} ICMP header

} Original IP header

} 8 byte of IP data

ICMP Source Quench (SQ) Packet.



Parameter time error (Parameter problem)

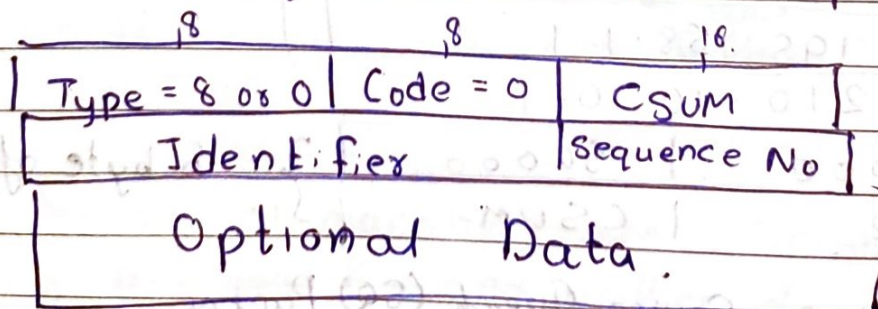
timeout error

Fragment error

* ICMP Query Messages

- How ICMP can be used to query network?
- In Query messages "rest of the header" is not all zero.

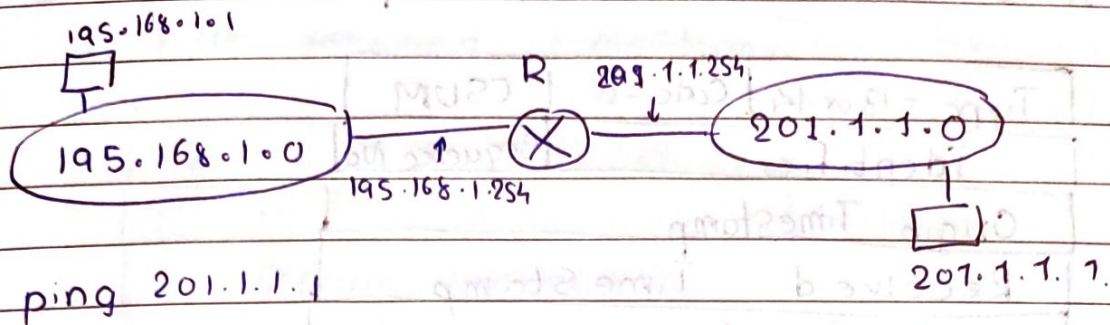
1. Echo Request and Echo Reply Message



Type = 8 → request
= 0 → Reply

• Identifier is usually process Id

- It is used to check communication between two IP



ICMP Message Packet

Type = 8	0	CSUM
PID	1	

4	5	00	28
101	0	000	0
2	1	1	HCSUM
195.168.1.1			
201.1.1.1			
ICMP Packet			
IP Packet			

PRE	SFD	DPA	SPA	T/L	Payload	CRC
-----	-----	-----	-----	-----	---------	-----

Ethernet Frame

When the packet reaches 201.1.1.1. The IP checks the protocol field to be 1, So it is ICMP packet

- IP forwards the ICMP packet to ICMP
- ICMP of 201.1.1.1 write a reply to 195.168.1.1

- Type = 0

- Source IP $\leftrightarrow$ dest IP

- dest will be same.

2. Timestamp Request and Reply

- It is used to find roundtrip time (RTT)

Type = 13 or 14	Code = 0	CSUM
Identifier		Sequence No
Origin Timestamp		
Received Time Stamp		
Transmit Time		

Origin: When request is generated.

Received: When request is received

Transmit: When reply is transmitted.

- ICMP generates a req at 1:30:12 PM
 $\Rightarrow 13 \times 60 \times 60 \times 1000 + 30 \times 60 \times 1000 + 12 \times 1000$
 $= \text{Origin Time Stamp.}$

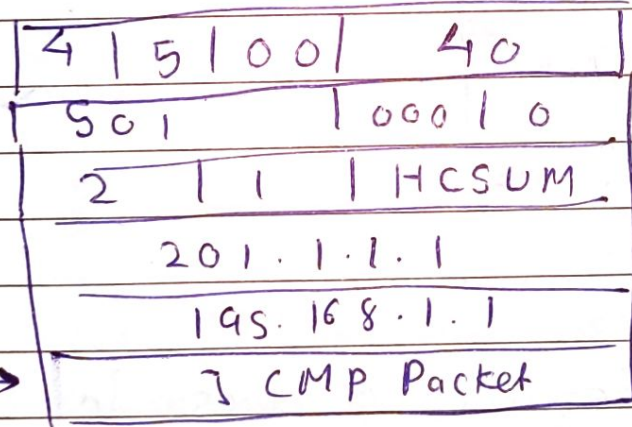
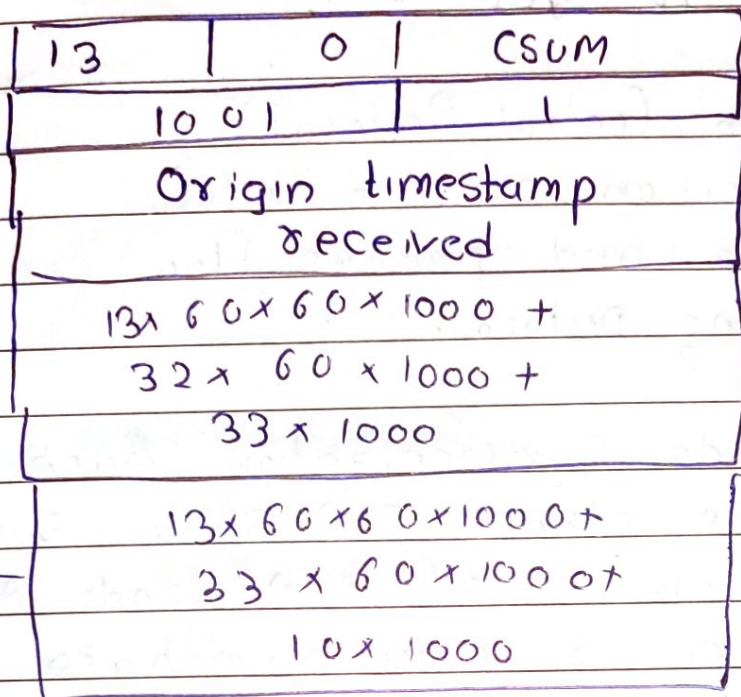
13	0	C	SUM
10011		1	
$13 \times 60 \times 60 \times 1000 +$ $30 \times 60 \times 1000 +$ 12×1000			

← Origin TimeStamp.

4	15	00	40
501	100010		
2	1	HCSUM	
198.168.1.1			
201.1.1.1			
ICMP Packet			

PRE|SFD|DPA|SPA|T/LI Payload|ERE

- Now, destination ICMP will receive the packet
- It fills received timestamp.



[PRE | SFD | DPA | SPA | T/L | Payload | CRC]

Ethernet Frame.

195.168.1.1

201.1.1.1

①

② Received

②

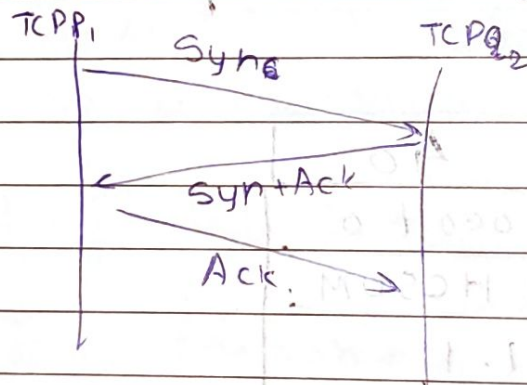
③ Transmitted.

- Now Source calculates difference between origin and received timestamp and ~~reserved~~ ^{transmitted} timestamp and ~~to~~ received and adds to get RTT.

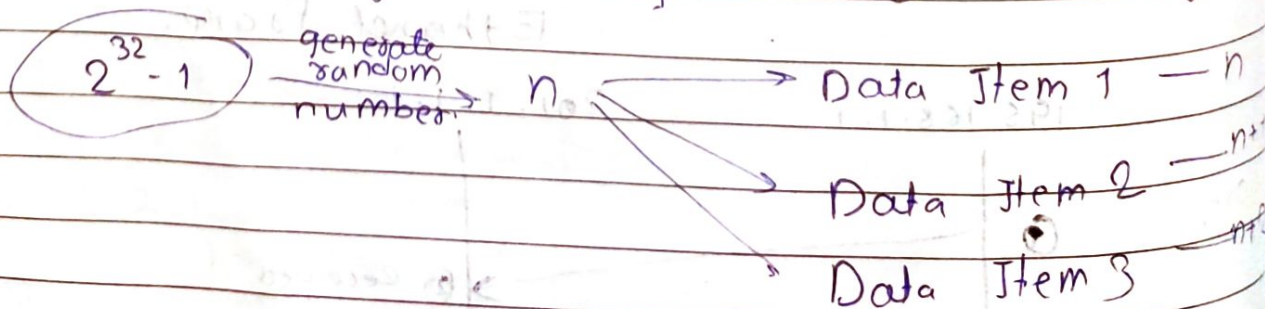
★ TCP (Transmission Control Protocol)

- It is a connection oriented protocol.
- It is reliable and provides flow control.
- It is Streaming protocol.

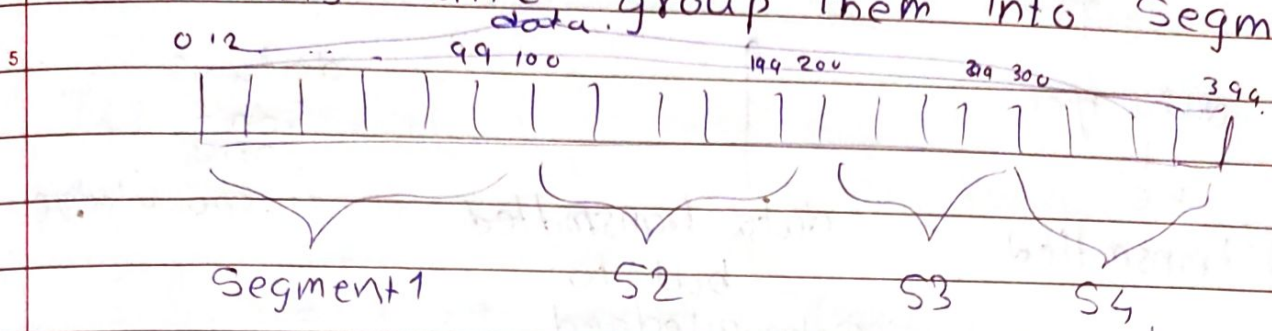
- First TCP Sends Synchronization packet to which the other TCP Sends Sync+Ack back to which initial TCP Sends Ack to establish a 3-way handshake connection.



- Stream is a Sequence of bytes. In TCP each byte is numbered and placed in Segments.
- This process of numbering is called Streaming



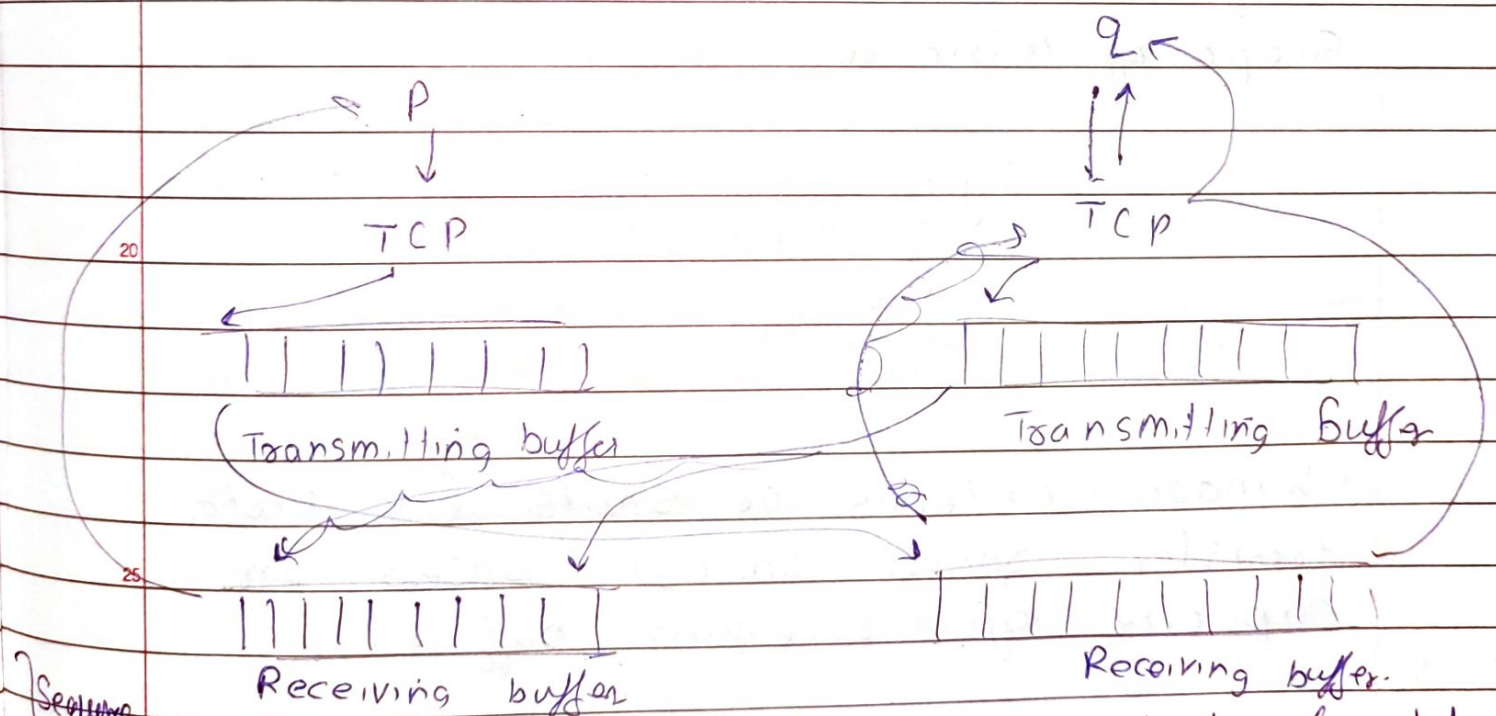
- When a process P generates data and sends the data to TCP, TCP numbers those data items and group them into segments.



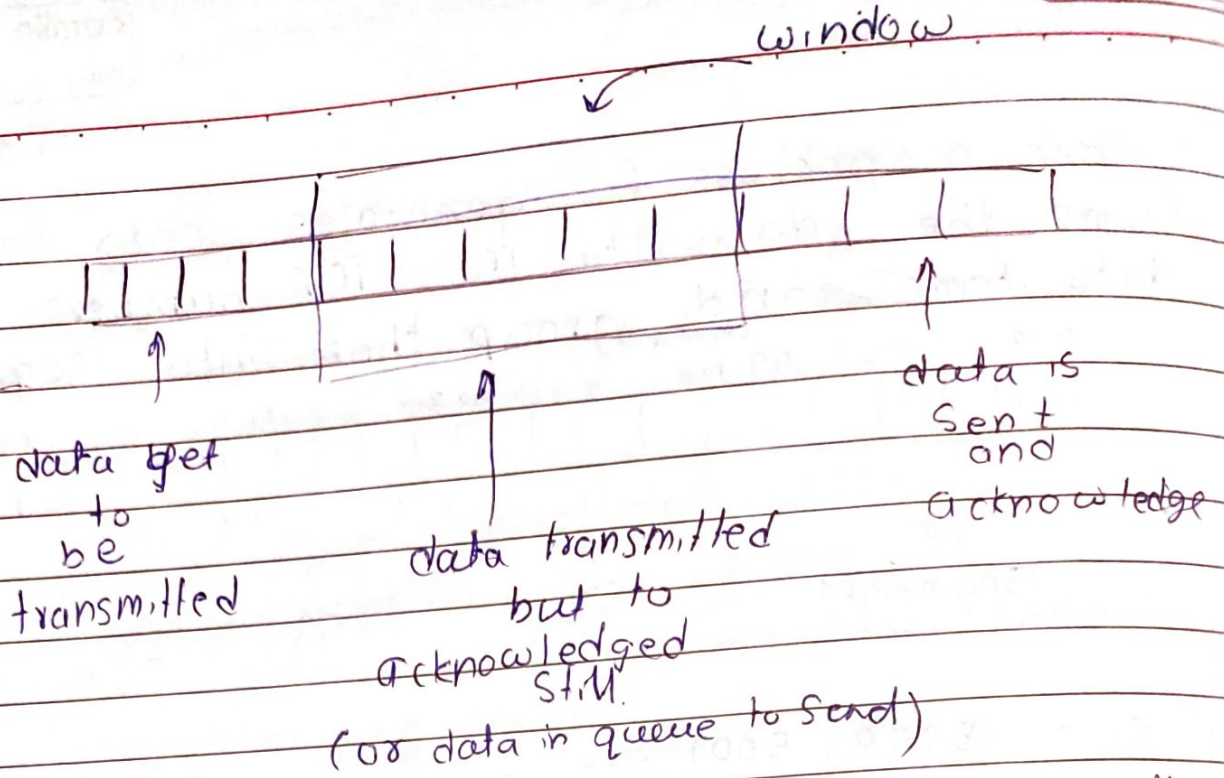
Segments of data.

S1 -	(5000, 5001	...	5099
S2 -	(5100, 5101		5199
S3 -	(5200, 5201		5299
S4 -	(5300, 5301		5399

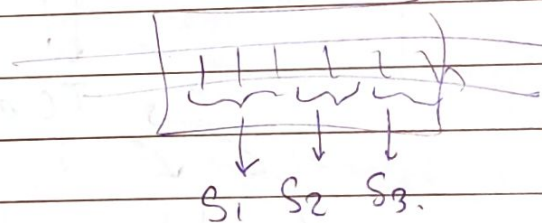
↑
Identification



TCP uses Sliding window protocol to transfer data. The sizes of transmission and receiving buffer may vary based on machine resources.



- As the ack for transmitted arrives the window keep on sliding.
- The Size of window is dependent on receiver's receiving buffer.
- Segments are formed for data in scope of window



- Window extends or shrinks (i.e. data transfer speed varies) based on capacity of receiving buffer.

Window can

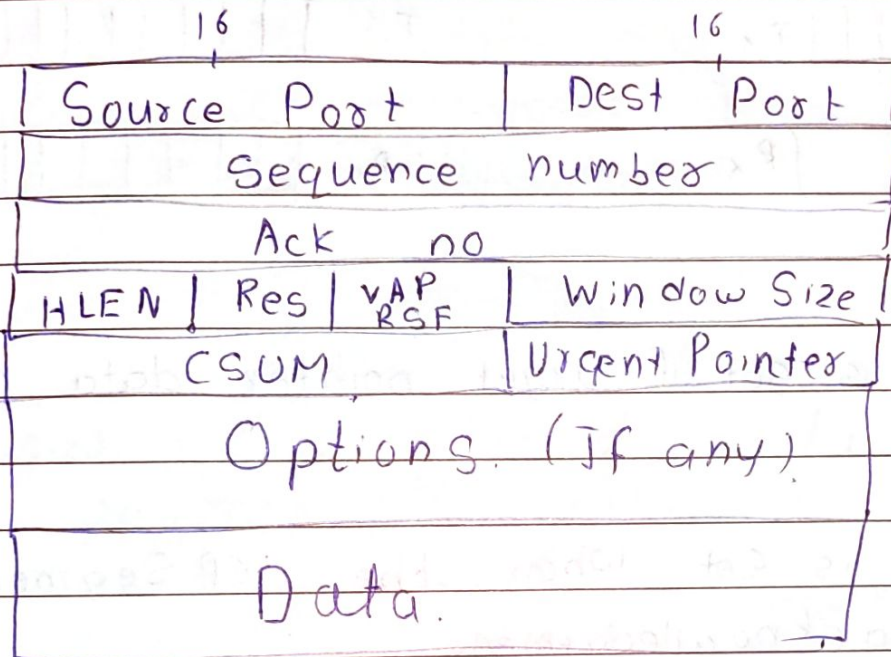
- Shrink
- Extend
- Slide.

This change of data rate (by shrinking and extending window) is called flow control.

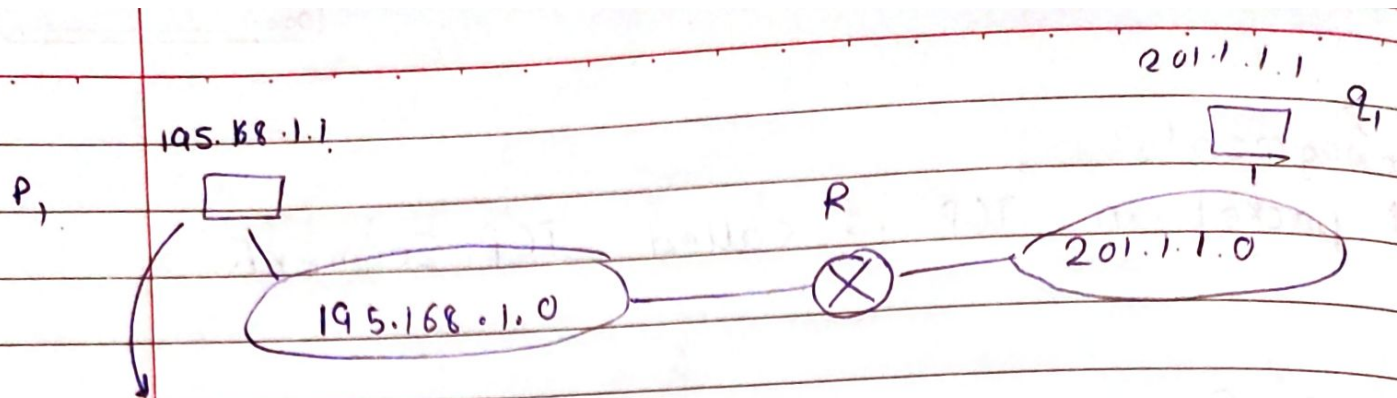
* Segments:

A packet in TCP is called TCP Segment.

* TCP Segment



- Sequence number is first byte of a Sequence of data items of a Segment.
- TCP believes in positive acknowledgement
Ack no is the Sequence no from which next data should be sent
All data before that number is received
- HLEN is length of header (It is specified in terms of 4 byte word). (without option values 5)
- Res (Reserves) is 6 bit field is all 0s
- UAP RST are flags which are 6 bit
 - URG RST
 - ACK SYN
 - PSH FIN



1. URG → It is set if urgent pointer data has value.

2. ACK → It is set when the TCP Segment is carrying acknowledgement.

3. PSH →

In certain circumstances TCP needs to Push data in those rare cases PSH is set.

4. RST → Set to reset connection

5. SYN → Synchronization flag is set which Sequence no is present.
(Segment is carrying Sequence no)

6. FIN → Finish flag is set to terminate connection

• Window Size is available buffer size of R_x (of ~~the~~ Source of this Segment)

• HCSUM - A pseudo header is added similar to UDP. For calculation of HCSUM.

SIP	195.168.1.1
DIP	201.1.1.1
all 0's	TCP segment length

Urgent ptr is offset of data that needs to be processed urgently.

$0 \dots \dots \dots 101$ ← Urgent ptr Value
 $0 \dots \dots \dots 101$

* Connection Setup:

TCP 1

ex. Read $n_1, n_2, \dots, n_{100}$

Call Socket (q1)

< 201.1.1.1, 60000 >

Get avg (through Socket)
display,

TCP 2

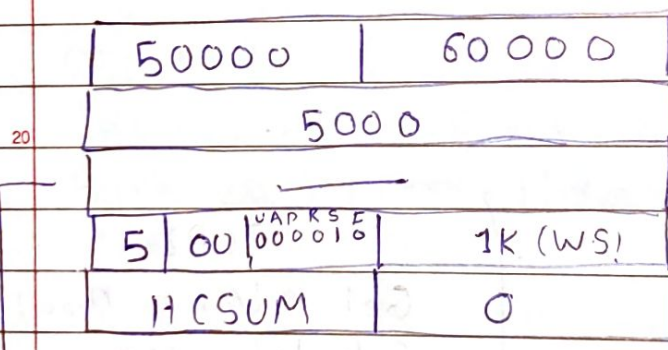
Get $n (n_1, \dots, n_{100})$

Calculate avg

write avg to Socket.

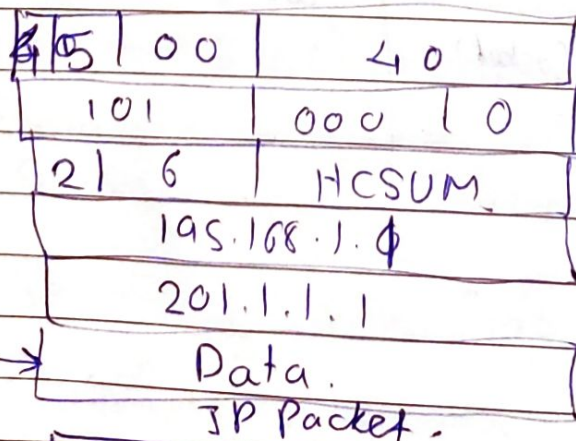
- Initially when TCP Socket is called after reading data (100 number's) a connection is established.
- First Before transmission of data, a connection is established first.
- For establishing connection initially ISN (Initial Sequence number) is generated randomly and exchange.
- Then MSS (Maximum Segment Size) is exchanged.
- Then Window Scale factor is negotiated (WSF)

All these values are negotiated while Synchronization



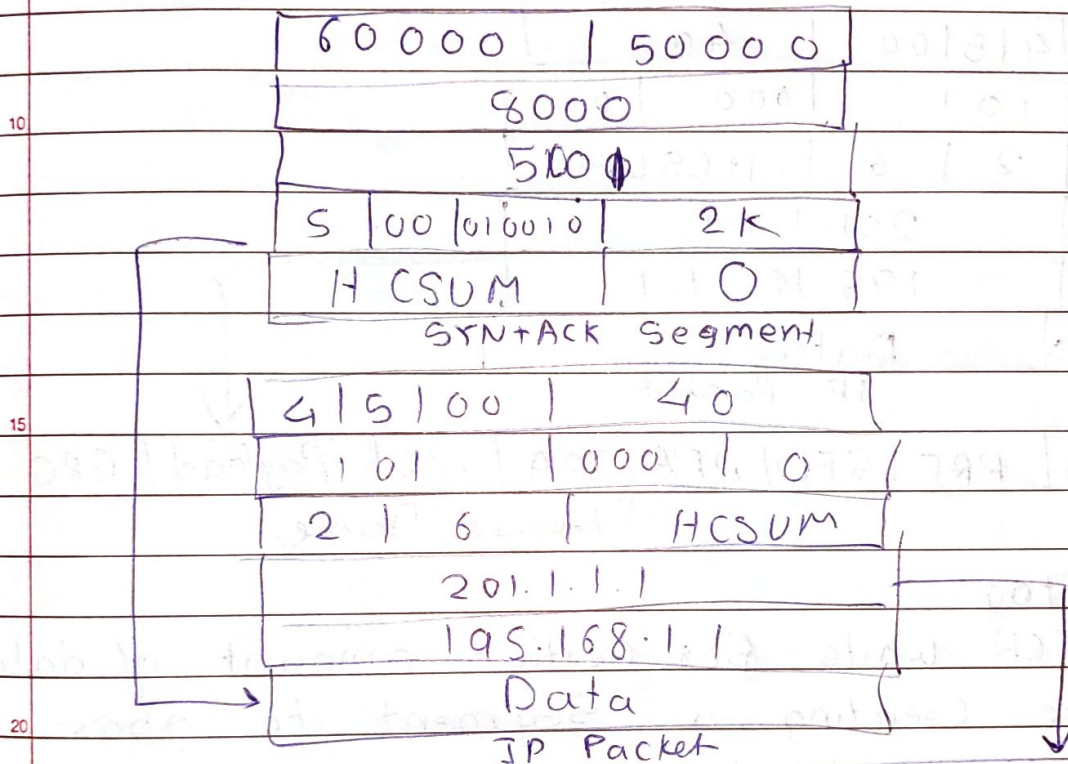
Syn Packet

MSS + WSF are added in Options
(HLEN is changed appropriately)



Ethernet Frame

- In a three-way handshake connection this is the first step
- The TCP2 now has Seq No, Max Segment, Size and WSF and Window Size of TCP1.
- The TCP2 now Sends Syn+Ack



PRE|SFD|DPA|SPA|T/4|Payload|CRC

Ethernet Frame.

- In some TCP implementation data transfer starts at Ack while other send null data.

50000	60000
5000	
8001	
51001010010	1k
HCSUM	0

Ack Segment

415100	40
101	000100
216	HCSUM
201.1.1.1	
195.168.1.1	
Data	

IP Packet

PRE	SFD	DPA	SPA	T/L	Payload	CRC
-----	-----	-----	-----	-----	---------	-----

Ethernet Frame

• PSH Flag:

- The TCP waits for certain amount of data before creating a segment to ~~avoid~~ avoid overhead.
- The push flag when set push data without waiting for more.

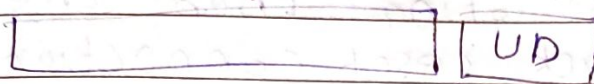
50000	12260000
5001	
8001	
51001001000	1k
HCSUM	0
cd xyz.	

Segment with push enabled

22-SSH Port.

Urgent Ptr:

If a Sender TCP Sends a chunk of data and if something goes wrong, if such cases certain urgent data like terminate etc needs to be sent.



xxxxxx

Normal data

offset

10

Urgent ptr value 10

Signifies that

urgent data is present till offset

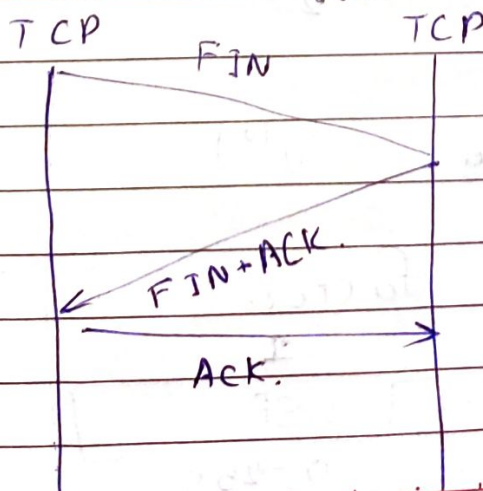
10 from which

further there is normal data.

50000	22
Seq No	
Ack no	
5/00/1.....	1k
1CSUM	10
Urgent data	Normal data

FIN Flag

It is used to termination of TCP connection.



Termination of connection

- RST Flag
- It is used to reset connection.

1. Situation 1: When TCP 1 Sends a connection request to TCP 2 and the port is not existing then the TCP 2 Sends back reset connection (RST Flag) back to TCP 1.

* TCP Options :

1. EOP (End of Options) - all 0's 00000000
Kind value is 0

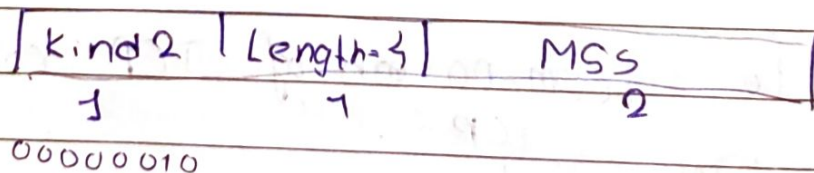
2. NOP (No Operation) - 00000001
Kind value 1

• It is used as filler option (to make options multiple of 4 bytes)

3. MSS (Maximum Segment Size)

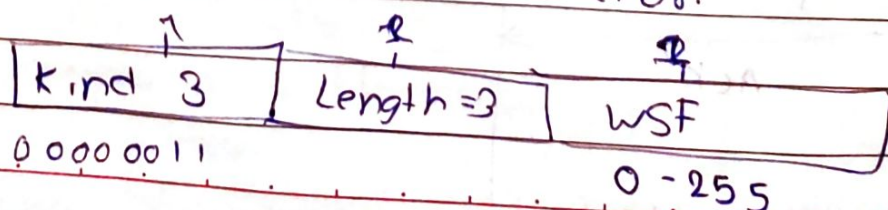
• Kind value is 2

• It is a 4 byte option.



Max Size : 65535 bytes (2^{16})

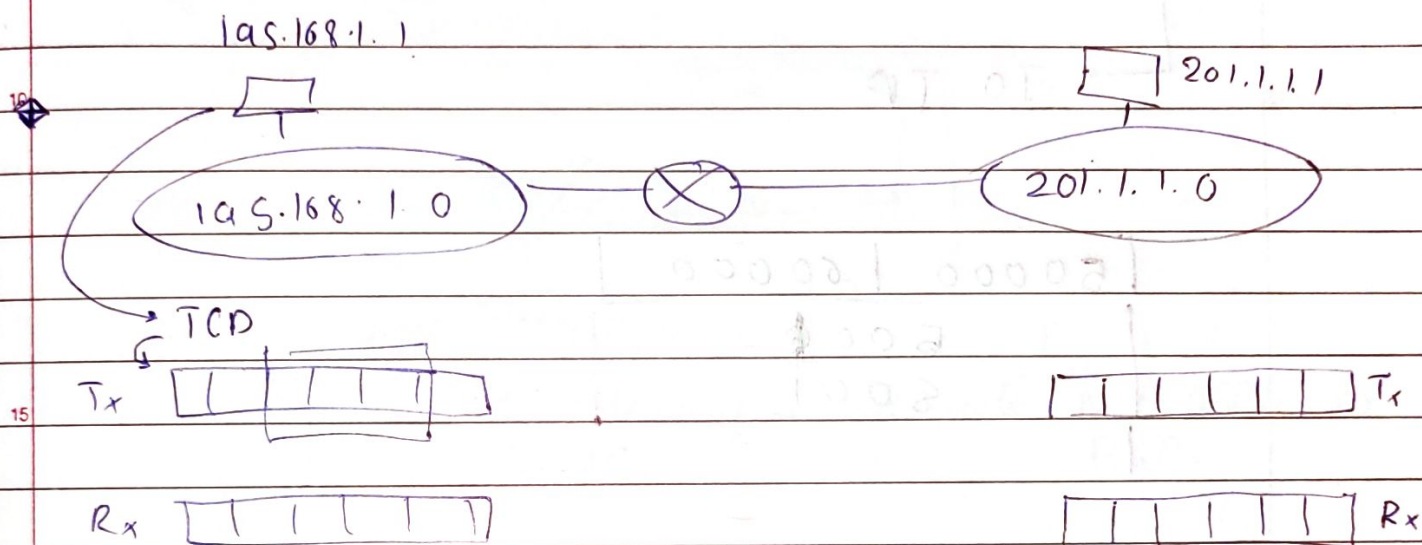
4. Window Scale factor.



$$\text{New Window Size} = \text{Window Size} \times 2^{\text{WSF}}$$

$$\text{Max New Size} = 2^{16} \times 2^{14}$$

Window Size WSF ← Permissible WSF is 14



50000	60000
5000	
—	
7	00001010
CS	
4	4
3	3

SYN

4	5	00	48
1	0	00	0
2	1	6	HCSUM
195.168.1.1			
201.1.1.1			

PRE | SFD | DPA | SPAT | L | Payload | CRE

60000	50000
8000	
5001	

SYN+ACK

7	0	010010	10k
---	---	--------	-----

CS		
2	4	12k
3	3	110

→ TO IP

50000	60000
-------	-------

5000
8001

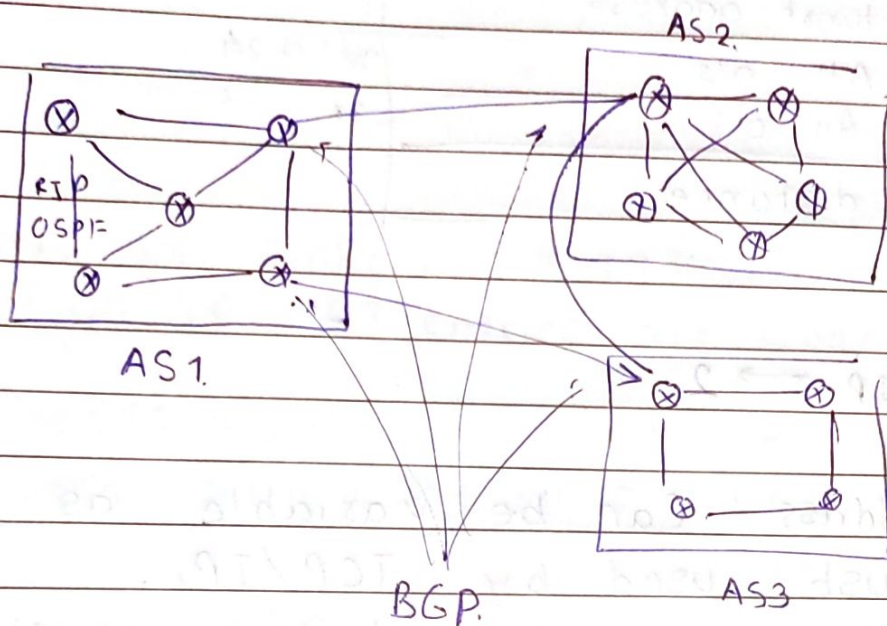
Seq no 5001 → 8001
 Ack no 8001 → 5101
 Seq no 5101 → 8100
 Ack no 8100 → 5201

* 25 Routing Protocols

RIP, OSPF, BGP

↓
 Routing Information protocol
 Open Shortest Path first
 Border Gateway protocol

1. Intradomain Routing → RIP, OSPF
 2. Interdomain Routing → BGP



(Border Gateway) → Use both RIP, OSPF or Routers BGP.

Interdomain Router.

* 20 RIP

It uses UDP well known port 520.

Router Advertisement

Router Advertises routing table.

- Command decides whether it is a request or response
 request → 1
 response / Advertise → 2.

- Version either 1 or 2 (RIP1, RIP2)

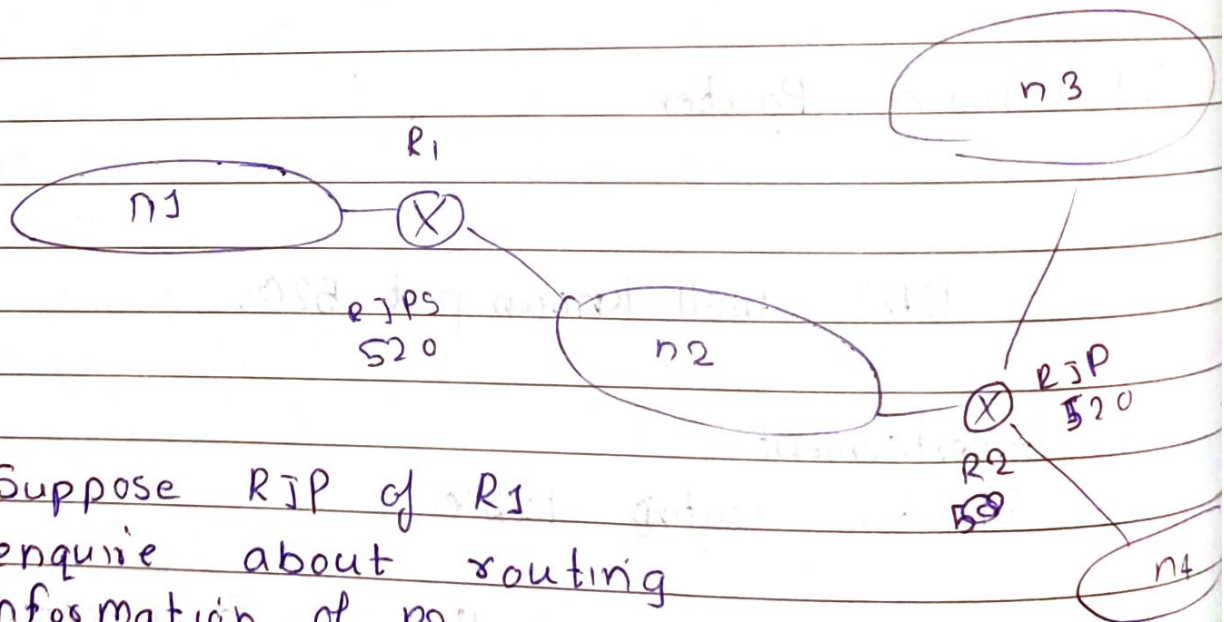
8	8	16		
Command family	Version	Unused all 0's	16	2
Network address			32	4
All 0's			32	4
All 0's			32	4
distance				

• Family:

For TCP/IP $\rightarrow 2$

• Network address: Can be variable as it is not just used by TCP/IP, TCP/IP just uses 4 bytes rest are 0's

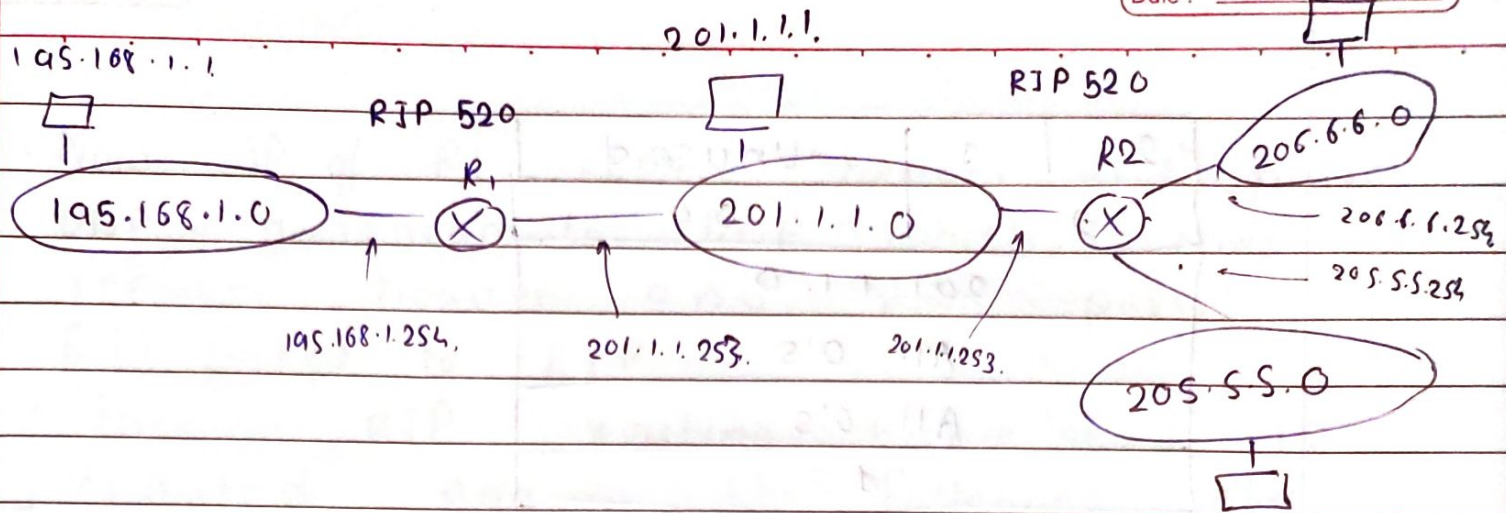
(Total K bytes)



Suppose RIP of R1 enquire about routing information of R2.

• In request distance is blank.

• If Network Address is all 0's in request then response has all network information (Request for All)



When we configure R1 (i.e. assign IP) default IP RT entries are added.

255.255.255.0	195.168.1.0	-	195.168.1.254
255.255.255.0	201.1.1.0	-	201.1.1.253

Similarly configuring R2 creates a default RT.

255.255.255.0	201.1.1.0	-	201.1.1.253
255.255.255.0	205.55.5.0	-	205.55.5.254
255.255.255.0	206.6.6.0	-	206.6.6.254

Suppose R2 makes an advertisement.

In this case destination port will be 520
(Every router advertised its RT periodically)
ex each 25-30 Sec.

2	1	Unused
2		00

201.1.1.0

All 0's

All 0's

1

2	1	Unused
2		00

205.5.5.0

All 0's

All 0's

1

2	1	Unused
2		00

206.6.6.0

All 0's

All 0's

1

RIP

UDP datagram

S Port	520
68	CS

RIP PKT

IP packet

4	5	00	88
Tot Len	100	10	
2	17	1	Checksum

201.1.1.253

255.255.255.255

UDP PKT

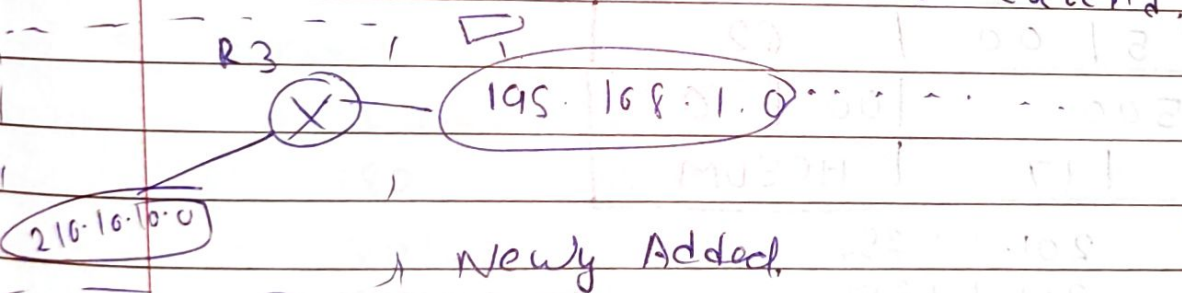
PRE | SFD | DPA | SPA | T/L | Payload | CRC

Now IP of R1 removes header and passes UDP datagram to UDP which further removes header and passes RIP packet to RIP

Through RIP routing table of R1 is updated adding the following fields.

hop count				
1	255.255.255.0	195.168.1.0	-	195.168.1.254
1	255.255.255.0	201.1.1.0	-	201.1.1.254
2	255.255.255.0	206.6.6.0	201.1.1.253	201.1.1.254
2	255.255.255.0	205.5.5.0	201.1.1.253	201.1.1.254

Now Suppose, we connect a router network to 195.168.1.0 to extend.



Suppose R1 generates a specific request to 205.5.5.0 to know the routing information

1	1	Unused
2	1	All 0's.
20S.S.S.0		
All 0		
All 0		
<blank>		

RIP Request

dynamic port	520
42	CSUM
Data	

4	5	00	62
500	00	0	
1	17	HCSUM	
201.1.1.254			
201.1.1.253			
Data			

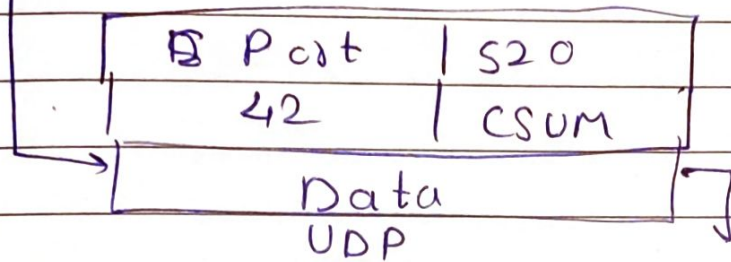
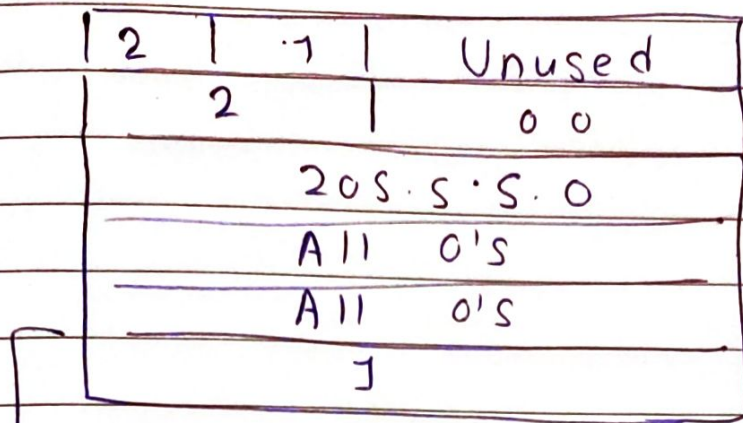
25S.25S.25S.0

PRE|SEF|DPA|SPA|T/L|Payload|CRC

* Timers in RIP:

- Each RIP has periodic timers set to 25-30 seconds which is used for periodic advertisement.
- One R.T is advertised, the value of timer is reset back.

* Expiration timer



To IP
 ↘ To ethernet

★ Timers

- The value of expiration timer is set to 180 Seconds
- The value in R.T are valid till 180 Seconds
- Once the timer hit's 0 the entries in R.T are removed if we don't get any update for those entries. (Unused)
- These entries are marked unused.

★ Garbage Collection timer

- It is set again to 120 Seconds
- It is used to remove entries from R.T which are marked unused.
- Entry is purged (removed)

- Periodic Timer — 25-30 Sec
- Expiration Timer — 180 Sec
- Garbage Collection time — 120 Sec